

(3 Hours)

[Total Marks : 80

N. B. : (1) Question No. 1 is compulsory.

(2) Attempt **any three** questions out of remaining **five** questions.

(3) Assumptions made should be clearly stated.

(4) Figures to the right **indicate full marks**.(5) Assume **suitable data** wherever **required** but justify the same.

1. (a) Give chomsky hierarchy of grammar with examples. 5
 (b) State and explain any 5 closure properties of regular languages. 5
 (c) Compare recursive and recursively enumerable languages. 5
 (d) State and prove equivalence of NFA and DFA. 5
2. (a) Design a DFA to accept strings over the alphabet set $\{a, b\}$ that begin with 'aa' but not end with 'aa'. 10
 (b) Convert $(0 + \epsilon) (1 0)^* (\epsilon + 1)$ into NFA with ϵ -moves and hence obtain a DFA. 10
3. (a) Design a MOORE and MEALY machine to decrement a binary number. 10
 (b) Give statement of pumping lemma for regular sets and hence prove that $\{w c w^R \mid w \in (a+b)^*\}$ is not regular where w^R is reverse of w . 10
4. (a) Obtain leftmost derivation, rightmost derivation and derivation tree for the string "cccbaccba". The grammar is 10

$$S \rightarrow SSa \mid SSb \mid c$$

 (b) Design Turing machine as generator to add two binary numbers and hence simulate for "110 + 10". 10
 Hint : Assume two way infinite tape.
5. (a) Design a PDA to accept language $\{a^{n-1} b^{2n+1} \mid n \geq 1\}$. 10
 (b) Convert the below given grammar to Chomsky Normal Form (CNF) and Griebach Normal Form (GNF) 10

$$E \rightarrow E+E \mid E * E \mid (E) \mid id$$

 Consider "id" as a single terminal/symbol.
6. (a) Design a Turing machine as acceptor for the language $\{a^n b^m \mid n, m \geq 0 \text{ and } m \geq n\}$. 10
 (b) Design PDA to check even parentheses over $\Sigma = \{0, 1\}$ 10